

## **Key Findings & Recommendations**

### **1. Best Model**

**XGBoost** (after GridSearchCV tuning) was selected as the best model based on the highest ROC-AUC score, reflecting its superior ability to distinguish churners from non-churners. The best model is selected by **ROC-AUC** because churn is imbalanced and ranking quality matters more than raw accuracy. Ensemble tree methods like XGBoost handle non-linear relationships and feature interactions well, making them well-suited for tabular business data.

### **2. Top 3 Factors Driving Churn (from XGBoost feature importance)**

- a) ***Tenure*** — Customers with very short tenure are far more likely to churn. New customers have not yet built loyalty or switching costs.
- b) ***Monthly Charges*** — Higher monthly charges correlate strongly with churn. Customers paying more are more price-sensitive and more likely to evaluate alternatives.
- c) ***Contract Type (Month-to-Month)*** — Month-to-month customers churn at ~42%, compared to <5% for two-year contracts. No contractual lock-in means zero friction to leave.

### **3. High Risk Customer Profile**

High-risk customers share a consistent profile:

- Short tenure (typically less than 12 months)
- High monthly charges (often greater than \$70/month)
- Month-to-month contracts with no long-term commitment
- Often without tech support or online security add-ons
- Frequently use fibre optic internet (which correlates with higher charges)

## 4. Actionable Business Recommendations

### ***Recommendation 1 — Targeted Retention Incentives for New High-Spend Customers:***

Deploy automated triggers for customers in months 1–6 who pay more than \$70/month on month-to-month contracts. Offer a discounted upgrade to a 1-year contract or a service bundle (TechSupport + OnlineSecurity) at a reduced price. This directly addresses the two biggest churn drivers simultaneously.

### ***Recommendation 2 — Contract Conversion Campaign:***

Run a proactive outreach campaign targeting all month-to-month customers flagged as High or Medium Risk. Offer a meaningful incentive (e.g., first two months free) to switch to a one-year contract. Two-year contract customers churn at <3%, demonstrating that commitment dramatically reduces churn.

## 5. Model Limitations & Improvements

- ***Class imbalance:*** ~26.5% of customers churned; without oversampling (e.g., SMOTE), the model may underperform on the minority class recall.
- ***No temporal features:*** If transaction-level data were available (support tickets, usage changes), sequential patterns could significantly improve predictions.
- ***Interpretability vs. performance trade-off:*** XGBoost is a black box; deploying SHAP values would make individual predictions explainable to business stakeholders.
- ***Static snapshot:*** This is a static historical dataset, so it does not capture changing customer behaviour over time. A real-world deployment would need periodic retraining as customer behaviour evolves.